

Attention-Based Parallel Multiscale Convolutional Neural Network for Visual Evoked Potentials EEG Classification

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Abstract—Electroencephalography (EEG) decoding is an important part of Visual Evoked Potentials-based Brain-Computer Interfaces (BCIs), which directly determines the performance of BCIs. However, long-time attention to repetitive visual stimuli could cause physical and psychological fatigue, resulting in weaker reliable response and stronger noise interference, which exacerbates the difficulty of Visual Evoked Potentials EEG decoding. In this state, subjects' attention could not be concentrated enough and the frequency response of their brains becomes less reliable. To solve these problems, we propose an attention-based parallel multiscale convolutional neural network (AMS-CNN). Specifically, the AMS-CNN first extract robust temporal representations via two parallel convolutional layers with small and large temporal filters respectively. Then, we employ two sequential convolution blocks for spatial fusion and temporal fusion to extract advanced feature representations. Further, we use attention mechanism to weight the features at different moments according to the output-related interest. Finally, we employ a full connected layer with softmax activation function for classification. Two fatigue datasets collected from our lab are implemented to validate the superior classification performance of the proposed method compared to the state-of-the-art methods. Analysis reveals the competitiveness of multiscale convolution and attention mechanism. These results suggest that the proposed framework is a promising solution to improving the decoding performance of Visual Evoked Potential BCIs.

Index Terms—Attention mechanism, brain-computer interface (BCI), convolutional neural network, fatigue, visual evoked potentials.

I. INTRODUCTION

BRAIN-COMPUTER Interfaces (BCIs) provide us a direct pathway between brain and computer. Fig. 1 shows the basic scheme of a general BCI system, which consists of data acquisition, data preprocessing, model training and prediction. The classification of EEG signals directly determines the performance of the system. Evoked potentials are robust EEG signals induced by sensory stimulation and they have been widely used in Brain Computer Interfaces (BCIs) [1], [2] due to their high recognition performance. Especially, Steady-State Visual Evoked Potential (SSVEP) [3], [4] is most commonly studied, which is the response in the visual cortex evoked by fixed frequency visual stimulus. However, continuous scintillation stimulation can cause physiological discomfort, leading to the reduction of prediction accuracy. Recently, research interest was turned to Steady State Motion Visual Evoked Potential (SSMVEP) [5], [6]. It obtains EEG signals from inductive motion rather than relying on high-intensity contrast, which can effectively alleviate the adverse effects of SSVEP. However, focusing attention to repetitive visual stimuli for a long time will cause physical and psychological fatigue, distracting the user's attention and reducing the classification performance [7], [8]. Therefore, for SSVEP- and SSMVEP-based BCIs, it is necessary to develop effective classification methods to improve the robustness of BCI systems, on both fatigue and normal states.

For SSVEP EEG decoding, it is important to accurately decode and analyze the frequency information. For example, power spectral density analysis (PSDA) is often used to identify peaks corresponding to the stimulus frequencies. Recently, a large number of frequency-based methods have been applied to the decoding of SSVEP. The most common method is the canonical correlation analysis (CCA) [9], a multivariable statistical method, which extends ordinary correlations to two sets of variables. It employs a template matching scheme between the EEG signals and the reference signals, and can improve the signal-to-noise ratio (SNR) in multiple channels. Various methods have been proposed to improve the basic CCA method for SSVEP. To reduce possible overfitting problem, especially

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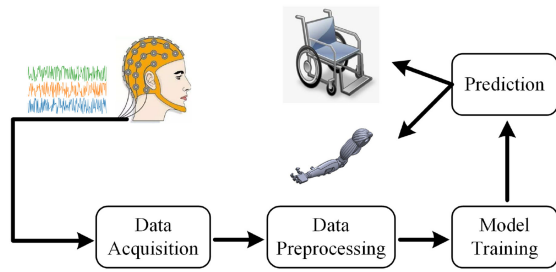


Fig. 1. The basic scheme of a general BCI system.

within a short time window, Zhang *et al.* [10] introduced an L1-regularized multiway canonical correlation analysis (L1-MCCA) for reference signal optimization. Chen *et al.* [11] proposed a filter bank canonical correlation analysis (FBCCA) to improve the recognition performance of SSVEP, which combines fundamental and harmonic frequency components. Vu *et al.* [12] applied deep CCA (DCCA) for frequency component extraction in SSVEP. They processed the input data through multi-layer neural networks to learn nonlinear transformations of the EEG signals and the reference signals before their correlations are computed. Zhang *et al.* [13] proposed a spectrum and phase adaptive CCA (SPACCA) for subject- and device-specific SSVEP-based BCIs. They designed a library of phase shifting reference signals and optimized a BCI system via the flexible reference signal generating approach. Nwachukwu *et al.* [14] proposed the individual template canonical correlation analysis (ITCCA), which incorporates features of the subjects' EEG data into the reference template for optimizing the reference signal. Similarly to SSVEP, SSMVEP signals also have significant frequency responses. These methods can thus be employed to analyze SSMVEP signals and achieve good performances. However, EEG signals in a fatigue state have relatively low SNR and inconspicuous classification features, hence it is a great challenge to decode the EEG signals of the fatigue state.

In order to learn robust feature representations, deep learning approach provides a flexible decoding method without the need of prior information, which has been widely used in many fields [15]–[21]. Particularly, the convolutional neural network has been applied to decoding EEG signals due to its rich feature representations [22]–[27]. Acharya *et al.* [28] employed the convolutional neural network for automated detection and diagnosis of seizure based on EEG signals. Chen *et al.* [29] proposed an EEG emotional feature learning and classification method using a deep convolution neural network. Gao *et al.* [30] proposed a novel EEG-based spatial-temporal convolutional neural network (ESTCNN) to detect driver fatigue. For SSVEPs and SSMVEPs, a common approach is to extract and fuse the frequency features. Kwak *et al.* [31] transformed SSVEP data using a fast Fourier transform (FFT) and then used a convolutional neural network for feature fusion and classification. To detect intentional control (IC) states and non-intentional control (NC) states in an SSMVEP-BCI system, Zhang *et al.* [32] proposed a novel convolutional neural network based on frequency-domain representation using FFT. These FFT-based approaches constrain the model to learn only spectral energy-based features, which may not be able to capture other task-relevant information.

Also, Waytowich *et al.* [33] proposed a compact convolutional neural network (Compact-CNN) for classification of SSVEP, which only requires raw EEG signals for automatic feature extraction. They used a large convolution kernel to extract robust frequency-domain features and decode SSVEP signals effectively. However, few studies focus on the classification of visual evoked potentials EEG under fatigue states. In this state, subjects' attention is not concentrated enough and the EEG signals have weaker reliable response.

In order to improve the accuracy of VEP signals under fatigue, we propose an attention-based parallel multiscale convolutional neural network (AMS-CNN). Multiscale convolution means convolutional layers with different filter sizes, which can extract robust features, such as the HS-CNN Dai *et al.* [34] proposed and CP-MixedNet Li *et al.* [35] proposed. For EEG signals, attention mechanism can calculate weights for different time segments, thereby helping the model to concentrate on the informative and important segments of a signal. Thus, many work introduce the attention mechanism to the classification of EEG signals [36]–[38]. Specifically, the AMS-CNN first extracts robust temporal representations via two parallel convolutional layers with small and large temporal filters respectively. Then, we employ two sequential convolution blocks for spatial fusion and temporal fusion to extract advanced feature representations. Further, we use attention mechanism to weight the features at different moments according to the output-related interest. Finally, we employ a full connected layer with softmax activation function for classification. Our proposed AMS-CNN is evaluated with 1s-long SSVEP and SSMVEP signals. The results show that our models achieve the best classification accuracy among comparable methods, especially on fatigue signals. Furthermore, we also show experimentally the competitiveness of multiscale convolution and attention mechanism. As a result, our proposed framework can provide an advanced solution for EEG decoding in BCI systems.

In summary, the main contributions of this paper are as follows:

- We propose a novel convolutional neural network framework to improve the classification performance of Visual Evoked Potentials EEG signals, especially under fatigue state.
- We adopt multiscale convolution to extract robust feature representation, due to the weak frequency response of VEP EEG signals in fatigue state.
- We employ attention mechanism to help the model focus on important segments of the signal. The subjects' attention is scattered under fatigue state, attention model weights it according to the features of each moment. For each feature, the attention module generated an independent group of values which would be used as weights.
- We experimentally demonstrate that the proposed model outperforms comparable methods on SSVEP fatigue dataset and SSMVEP fatigue dataset.

The rest of the paper is organized as follows: Section Table II describes the datasets for evaluation and the details of proposed AMS-CNN. Section Table III introduces comparable methods and presents experimental results. Section Table IV gives the conclusions.

II. METHODOLOGY

A. The Attention-Based Parallel Multiscale Convolutional Neural Network

Different brain activities may trigger changes in different brain regions in both time domain and frequency domain. For visual evoked potentials, the discriminative frequency response appears in the visual cortex. When the subject is fatigued, however, the frequency response will be weaker and the subject's attention will be distracted. To solve these issues, we proposed the attention-based parallel multiscale convolutional neural network (AMS-CNN). Particularly, we first extract robust temporal representations via two parallel convolutional layers with small and large temporal filters respectively. Then, we employ two sequential convolution blocks for spatial fusion and temporal fusion to extract advanced feature representations. Further, we use attention mechanism to weight the features at different moments according to the output-related interest. Finally, we employ a full connected layer with softmax activation function for classification. Fig. 2 shows the architecture of AMS-CNN.

1) **Feature Extraction:** To extract robust temporal representations from raw EEG signals, the feature extraction part consists of two parallel convolutional layers with small and large filter sizes. The small kernel could better capture the temporal information while the large kernel could better capture the frequency information [38]. Here, the large filter size is the same as the sampling frequency and the small filter size is set as 16 which is chosen by cross validation. Note that, the strides of these two convolutional layers are set as (1×1) and the padding mode is set as SAME, which means padding 0 around the margin of the input data. Therefore, the output feature maps of the two convolutional layers have the same size. Then, we apply a linear activation function and concatenate these two kind of feature maps for subsequent fusion. Given a multi-channel input X , where $X \in \mathbb{R}^{C \times T}$, C means the number of channels and T means the number of discretized time points, the output of the feature extraction part is

$$B_1 = \text{Cat}([f(\text{Conv}(X, \text{kenel}_c^1)), f(\text{Conv}(X, \text{kenel}_c^2))]) \quad (1)$$

where $\text{Cat}(\ast)$ is the concatenating function, $f(\ast)$ is the activation function, $\text{Conv}(\ast)$ is the regular convolution, kenel_c^1 represents the small kernel and kenel_c^2 represents the large kernel.

2) **Feature Fusion:** In feature fusion part, we employ two sequential convolution blocks to fuse multichannel features in spatial and temporal dimensions. Each CNN block performs five operations sequentially: convolution, batch normalization, activation of the exponential linear unit (ELU), average pooling and dropout. As shown in Fig. 2, each *Conv* block shows the number of spatial filters to learn per temporal filter, the form of convolution and the kernel size. Each *Pool* block shows the form of pooling and the pooling size, and the stride size is same as pooling size. Particularly, we use depthwise convolution, a part of the depthwise separable convolution, for spatial fusion in the first CNN and temporal fusion in the second CNN. Differing from standard convolution, here the depthwise separable

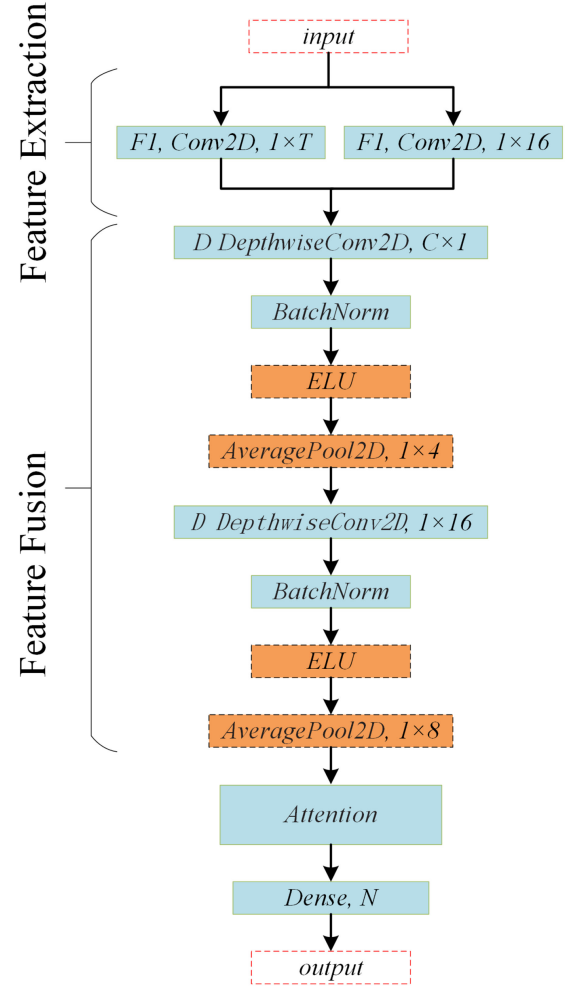


Fig. 2. The architecture of the AMS-CNN, where indigo rectangle is the trainable layer and orange rectangle is the non-trainable layer. $F1$ = number of temporal filters, T = number of time points, C = number of channels, D = number of spatial filters to learn per temporal filter and N = number of classes.

convolution consists of two steps: (1) a spatial feature learning step (depthwise convolution) and (2) a channel combination step (pointwise convolution). The depthwise convolution performs regular convolution within each depth slice and the pointwise convolution fuses the information across the whole depth using a 1×1 kernel. The depthwise convolution is

$$\text{Depthwise-Conv}(X) = \text{Cat}([\text{Conv}(X^1, \text{kenel}_d^1), \text{Conv}(X^2, \text{kenel}_d^2), \dots, \text{Conv}(X^l, \text{kenel}_d^l)]) \quad (2)$$

where l is the depth of X , X^i is the i th depth slice of X and kenel_d^i is the kernel corresponding to X^i . The depthwise convolution can reduce the number of parameters and explicitly decouple the relationship between input and output in convolution operations. In our models, depthwise convolutions with one spatial filter are employed and the output of depthwise convolutions has the same size as input.

3) **Attention Mechanism:** Based on the output of feature fusion, we use attention mechanism to weight the features at different time steps. Fig. 3 shows the attention model used in AMS-CNN, which is based on the similar principle as the one

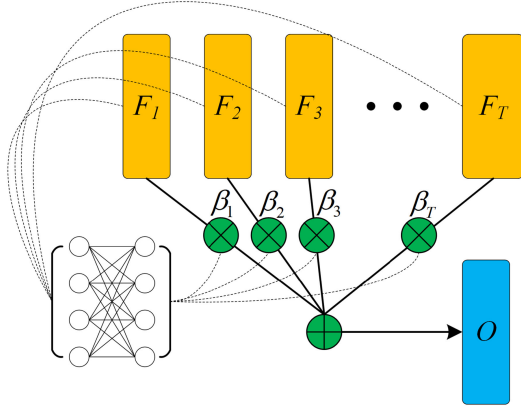


Fig. 3. The structure of attention module. The feature vector at each time step F_t is fed into a fully connected layer to produce a probability vector β_t , where $t = 1, 2, \dots, T$ and T is the total number of time steps. The output vector O is the weighted average of F_t with weighting given by β_t .

used in [39]. In the attention module, one fully connected layer along with a hyperbolic tangent function is applied to the feature vector at each time step. The output size of fully connected layer is the same as the input feature vector. Softmax is then employed for each series, so that the values in each series summed to 1. For each feature, the attention module generated an independent group of values which would be used as weights. At last, the weighted average of input is computed as the output of the attention model. Given a model which produces a feature vector $F_t \in \mathbb{R}^{d \times 1}$ at each time step t , attention model compute a output vector $O \in \mathbb{R}^{d \times 1}$ as the weighted mean of the feature vector F by

$$O = \sum_{t=1}^T \beta_t \odot F_t \quad (3)$$

where T is the total number of time steps, $\beta_t \in \mathbb{R}^{d \times 1}$ is the weight computed at each time step t based on F_t and $\odot$ represents element-wise multiply. The weight β_t can be formulated as follows:

$$h_t = \tanh(WF_t + b) \quad (4)$$

$$\beta_t = \frac{\exp(h_t)}{\sum_{k=1}^T \exp(h_k)} \quad (5)$$

where $W \in \mathbb{R}^{d \times d}$, $b \in \mathbb{R}^d$ are the trainable neural network parameters. By introducing the attention mechanism as above, we weight the time step separately for each feature. It can help the model focus on important segments of the signal, thereby improving the classification performance.

B. Model Training

The AMS-CNN is trained using a categorical cross-entropy loss function of the form

$$\text{loss} = \sum_{i,j} y_{i,j} \log(\hat{y}_{i,j}) \quad (6)$$

where $y_{i,j}$ are the true labels, $\hat{y}_{i,j}$ are the prediction values, i denotes the sample number and j denotes the class. All models

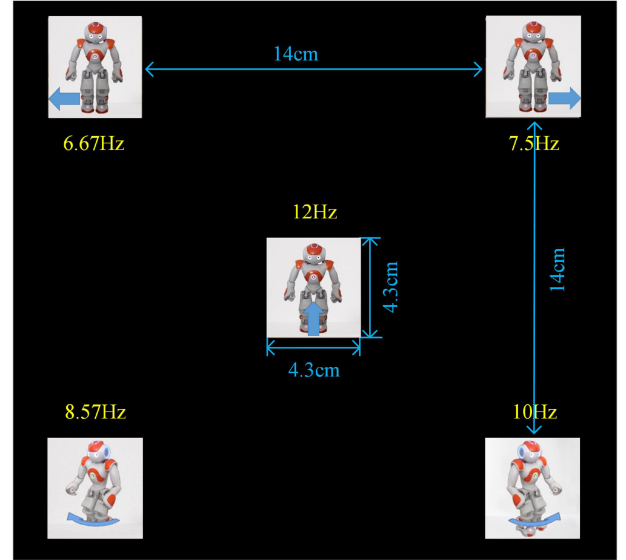


Fig. 4. The visual stimulator of the SSVEP fatigue experiment, where yellow words mean the stimulus frequencies and indigo words mean the measures of the stimulator.

are trained for 300 epochs and the mini-batch size is 64. We use the Adam optimizer with default parameters for parameter adjustment. The learning rate is set as 0.001. The dropout probability is set as 0.5 for all dropout layers. In our models, the number of temporal-spatial filter pairs is set as 64 ($F1 = 32$ and $D = 1$). During the process of model optimization, we save the optimal model that has the lowest loss in the validate set and assess the classification performance through the test set. This training process is implemented on a workstation equipped an Intel CPU (i7-7700K, 4.2 GHz) and an NVIDIA GPU (GTX 1060). The Keras library, extended from Tensorflow, is used to build the models.

III. RESULTS

A. Datasets

As shown in Table I, the datasets consist of two parts:

1) **Dataset a:** Dataset A (SSVEP fatigue dataset) was collected by our lab based on the SSVEP fatigue experiment. It was recorded from 9 subjects (5 female and 4 male students, average age: 24.0 ± 1.0 years old) in their normal states and fatigue states. The SSVEP visual stimulator is shown in Fig. 4. Five stimulus frequencies of 6.67 Hz, 7.50 Hz, 8.75 Hz, 10 Hz and 12 Hz were selected. Our experiments consist of several sessions, each including ten trials. Before each session, all subjects need to fill out a fatigue questionnaire to measure their fatigue levels, which contains 10 questions relating to general physical fatigue and 6 questions relating to mental fatigue according to the Chalder Fatigue Scale (CFS). All subjects participate the experiments in normal and fatigue state, respectively, and each experiment consists of 3 sessions. So, we obtained 150 trials (30 trials per stimulus frequency) of 4s in total under normal and fatigue state, respectively. The EEG was recorded via a 40-channel NuAmps amplifier of Neuroscan Company and the sampling frequency was set as 1 KHz.

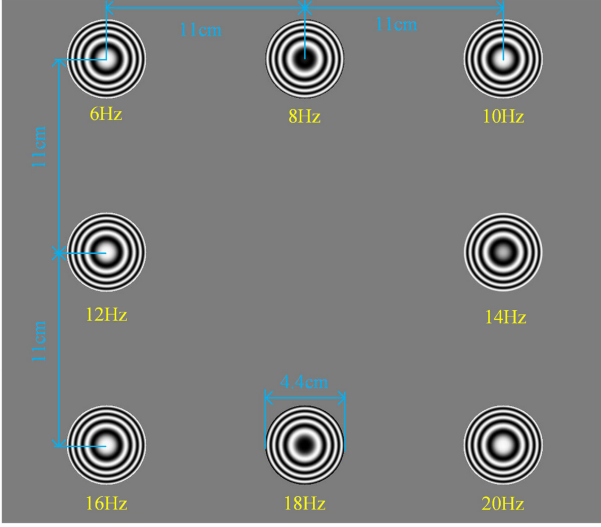


Fig. 5. The visual stimulator of the SSMVEP fatigue experiment, where yellow words mean the stimulus frequencies and indigo words mean the measures of the stimulator.

2) **Dataset B:** Dataset B (SSMVEP fatigue dataset) was also collected by our lab based on the SSMVEP fatigue experiment. It was recorded from 10 subjects (5 female and 5 male students, average age: 24.0 ± 1.0 years old) in their normal states and fatigue states. We used Newton's ring motion setting, which composes of the same black and white rings of 4 groups. The SSMVEP visual stimulator is shown in Fig. 5. We selected 8 stimulus frequencies: 6 Hz, 8 Hz, 10 Hz, 12 Hz, 14 Hz, 16 Hz, 18 Hz and 20 Hz. Similarly to Dataset A, the experiment has a total of 240 trials of 4s lengths under normal and fatigue state, respectively. The EEG was recorded via a 64-channel SynAmps2 amplifier of Neuroscan Company and the sampling frequency was set as 1 KHz.

For the recorded EEG signals, we employ the Independent Component Analysis (ICA) method to remove eye movement and blink artifacts based on EEGLAB [40]. Then, we obtain 30-channel data of SSVEP and 62-channel data of SSMVEP. To reduce the calculation complexity, we choose 12 channels (PO7, PO5, PO3, POZ, PO4, PO6, PO8, CB1, O1, OZ, O2, CB2) as the final input for SSMVEP and choose 8 channels (T5, P3, PZ, P4, T6, O1, OZ, O2) as the final input for SSVEP. These EEG channels are in the occipital region and parietal occipital region, which have commonly been selected for VEP-based BCI studies [41]. Deep neural networks require a large number of samples for training. However, each subject can only offer a small number of trials to train the model and each trial lasts for only several seconds. So, we chose cropping to expand the number of samples. For each subject, we selected 80% trials for training, 10% trails for validating and 10% trails for testing. For each trial, we discarded the first second, in which the signals may contain too much artifacts due to the reaction time of subjects. Then, the signals of training set and validating set were split into segments of length 1s via a sliding window, and the sliding step is 0.25s. The signals of testing set were split into segments of length 1s without overlapping. At last,

the training set and validating set were expanded to 9 times, while the testing set were expanded to 3 times. All data were bandpass filtered using a Butterworth filter between 5 and 25 Hz for SSVEP, and between 5 and 40 Hz for SSMVEP. The filter band is related to the range of the stimulation frequencies. Then, all data were downsampled to 256 Hz and standardized for subsequent analysis. The standardization method is described by

$$y' = \frac{y - y_{mean}}{y_{std}} \quad (7)$$

where y' and y are the signals before and after the standardization, y_{mean} and y_{std} are the mean and standard deviation of y .

B. Comparable Methods

To validate the performance of our proposed framework, we make a comparison with other common methods on the above two datasets. Here, we briefly introduce these comparable models.

1) **Standard CCA:** Canonical correlation analysis (CCA) is a multivariate statistical analysis method that finds the global correlation between two multi-dimensional variables. Given two multi-dimensional variables X and Y , CCA aims to find projection vectors A and B , so that the two linear projections $x = X^T A$ and $y = Y^T B$ have a maximal correlation coefficient. For SSVEP and SSMVEP, CCA is employed to compute the correlation between multi-channel EEG signals and a set of reference signals. Reference signals Y_k consist of sines and cosines corresponding to the fundamental and harmonic frequencies of the k th stimulus frequency.

$$Y_k = \begin{pmatrix} \sin(2\pi f_k t) \\ \cos(2\pi f_k t) \\ \vdots \\ \sin(2\pi N f_k t) \\ \cos(2\pi N f_k t) \end{pmatrix} \quad (8)$$

where f_k is the k th stimulus frequency and N is the number of harmonics. In this work, we choose three harmonics for each stimulus frequency. The correlation coefficients between EEG signals and each set of reference signals are computed and the maximum-value method is employed for classification. $f_s = \max_f \rho(f_k), k = 1, 2, \dots, K$, where $\rho(f_k)$ means the CCA coefficient of EEG signal X and reference signal Y_k , K is the number of categories ($K = 5$ for SSVEP and $K = 8$ for SSMVEP).

2) **CNN Based on FFT:** Due to the significant frequency response of the VEP signals, many works analyze signals from the frequency domain. Convolutional neural network based on FFT consists of two steps: the transformation from time domain to frequency domain via FFT and the decoding using CNN. In this work, we first transform the data by fast Fourier transform (FFT) and choose 60 samples from each channel, corresponding to 0–60 Hz. Then, we use a five-layers CNN for decoding the signals, which consists of a spatial convolutional layer, two sequential frequency convolutional layers and two dense layers. Table II shows the details of the CNN structure based on FFT. To

TABLE I

DATASETS PROPERTIES, WHERE FS MEANS SAMPLING FREQUENCY

Name	Source	Subjects	Categories	Channels	fs (Hz)
Dataset A	experiment	9	5	40	1000
Dataset B	experiment	10	8	64	1000

simplify the expression, we call the CNN based on FFT method as FFT+CNN below.

3) **Compact-CNN**: Compact convolutional neural network (Compact-CNN), known as EEGNet, is proposed for classifying raw EEG signals [33]. The compact-CNN consists of three convolutional blocks. The first layer of the network employs temporal convolutions as bandpass filters. Then, depthwise convolutions are used as spatial filters to reduce the dimensionality of the data and separable convolutions are used to further combine information across filters. After each convolution layer, there are batch normalization, 2D average pooling, and dropout layers. The compact-CNN is detailed in Table III.

4) **ALS-CNN & ASS-CNN**: ALS-CNN is implemented by replacing small kernels with large kernels while ASS-CNN is implemented by replacing large kernels with small kernels in feature extraction part of AMS-CNN. Therefore, they can only learn the single-scale temporal representation.

C. Overall Comparison

To make an effective comparison, we apply the same training method for all models and the 10-fold cross-validation is employed. We consider the average classification accuracy as the within-subject evaluation indicator. We evaluate the classification performance via average accuracy (ACC) and variance (VAR) of the differences across subjects. Table IV shows the average accuracy and variance of different methods on the SSVEP dataset and the SSMVEP dataset. Obviously, the classification accuracy will be significantly reduced when the subject is fatigued. While, our proposed method achieve the best classification performance among the comparable methods, whether the subjects are normal or fatigued. For SSVEP, the mean classification accuracy of AMS-CNN reach 96.42% on normal signals and 77.56% on fatigue signals, which outperform standard CCA by over 20%. For SSMVEP, the mean classification accuracy of AMS-CNN can also reach 78.11% on normal signals and 54.24% on fatigue signals, which outperform standard CCA by around 30%. Note that, our method is more effective in improving the classification performance of EEG signals under fatigue, due to multiscale convolution and attention mechanism. What's more, AMS-CNN receives the smaller variance comparing with other methods except CCA, indicating that our method effectively improves the classification accuracy of the subjects with unobvious characteristics and weakens the classification differences between subjects.

When the subjects are under fatigue state, the mean accuracy of SSMVEP can only reach 54.24%, the problem mainly comes from two aspects. On one side, we asked subject 4, 5, 9 and found that they were too tired during the experiment to complete the experiment well, resulting in a lower mean classification

TABLE II

DETAILS OF CNN STRUCTURE BASED ON FFT, WHERE C MEANS THE NUMBER OF CHANNELS AND N MEANS THE NUMBER OF CLASSES

Layer	Filters	Size	Output	Activation	Options
Input			(C, 60, 1)		
Conv2D	8	(C, 1)	(1, 60, 8)	Linear	Mode = valid
Conv2D	16	(1, 3)	(1, 58, 16)	Linear	Mode = valid
Conv2D	16	(1, 3)	(1, 56, 16)	Linear	Mode = valid
Batch Norm			(1, 56, 16)		
Activation			(1, 56, 16)	Elu	
Flatten			(896)		
Dropout			(896)		Rate=0.5
Dense		32	(32)	Relu	
Dense		N	(N)	Softmax	

TABLE III

DETAILS OF COMPACT-CNN STRUCTURE, WHERE C MEANS THE NUMBER OF CHANNELS AND N MEANS THE NUMBER OF CLASSES

Layer	Filters	Size	Output	Activation	Options
Input			(C, 256, 1)		
Conv2D	64	(1, 256)	(C, 256, 64)	Linear	Mode = same
Batch Norm			(C, 256, 64)		
Depthwise Conv2D	64	(C, 1)	(1, 256, 64)	Linear	Mode = valid
Batch Norm			(1, 256, 64)		
Activation			(1, 256, 64)	Elu	
Average Pool2D		(1, 4)	(1, 64, 64)		
Dropout			(1, 64, 64)		Rate=0.5
Separable Conv2D	64	(1, 16)	(1, 64, 64)	Linear	Mode = same
Batch Norm			(1, 64, 64)		
Activation			(1, 64, 64)	Elu	
Average Pool2D		(1, 8)	(1, 8, 64)		
Dropout			(1, 8, 64)		Rate=0.5
Flatten			(512)		
Dense		N	(N)	Softmax	

accuracy of fatigue state. On the other side, we use a short time window for classification, aiming to achieving a high execution efficiency of BCI systems. The short time window will cause a decrease in classification accuracy. However, some BCI applications, such as smart home systems, are based on weak classification control, which allows producing a result after multiple cycles. Thus, subjects can use this type of BCI system normally after using our method for improvement even if their classification accuracy is lower.

When we only use large convolution kernels or small convolution kernels, it will lead to a decrease in accuracy, which demonstrates the effectiveness of multiscale convolution. As shown in Table IV, the accuracy of AMS-CNN and ALS-CNN is significantly higher than that of ASS-CNN. Compared with ALS-CNN, the accuracy of AMS-CNN is slightly higher and the variance of AMS-CNN is smaller. These suggest that large kernels is better for extracting discriminative features than small kernel while multiscale convolution can improve the robustness of the model.

TABLE IV
THE OVERALL COMPARISON OF CLASSIFICATION PERFORMANCE

Dataset	State	Method	ACC (%)	VAR
SSVEP	Normal	CCA	76.08	0.0126
		FFT+CNN	86.67	0.0073
		Compact-CNN	96.20	0.0010
		ALS-CNN	96.57	0.0007
		ASS-CNN	93.63	0.0025
		AMS-CNN	96.42	0.0006
	Fatigue	CCA	48.80	0.0076
		FFT+CNN	60.05	0.0174
		Compact-CNN	74.57	0.0174
		ALS-CNN	76.99	0.0132
SSMVEP	Normal	ASS-CNN	73.73	0.0153
		AMS-CNN	77.56	0.0117
		CCA	40.19	0.0262
		FFT+CNN	56.90	0.0139
		Compact-CNN	76.32	0.0163
		ALS-CNN	77.71	0.0130
	Fatigue	ASS-CNN	72.11	0.0163
		AMS-CNN	78.11	0.0109
		CCA	29.14	0.0281
		FFT+CNN	34.96	0.0521

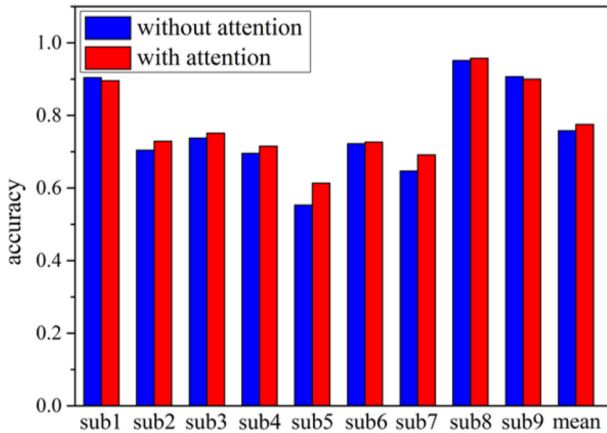


Fig. 6. The accuracy comparison of the AMS-CNN model without and with attention mechanism on SSVEP fatigue signals, where the blue color represents the model without attention mechanism and the red color represents the model with attention mechanism.

D. Influence of Attention Mechanism

In addition to multiscale convolution, the attention mechanism also affects the classification performance. Due to the obvious improvement of AMS-CNN on fatigue signals, in this section, we focus on the influence of attention mechanism on the classification performance under fatigue state. We implement the model without attention mechanism via removing attention model in AMS-CNN and directly feeding the features to dense layer. Fig. 6 and Fig. 7 show the accuracy comparison of the model without and with attention mechanism on SSVEP and SSMVEP fatigue datasets. We can observe that attention mechanism can improve the classification accuracy of SSVEP and SSMVEP signals, especially for the subjects with less obvious

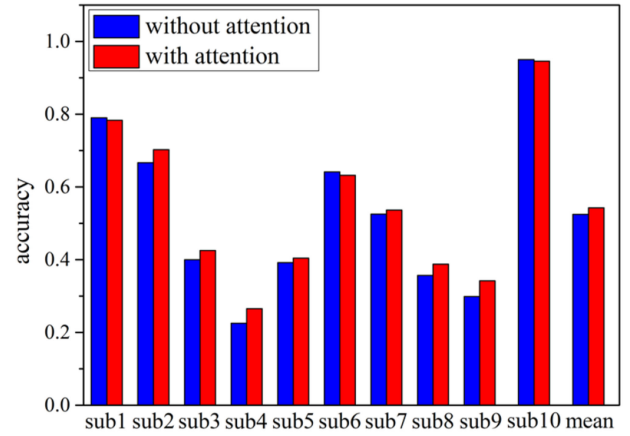


Fig. 7. The accuracy comparison of the AMS-CNN model without and with attention mechanism on SSMVEP fatigue signals, where the blue color represents the model without attention mechanism and the red color represents the model with attention mechanism.

characteristics. For example, for subject 5, 7 of SSVEP dataset and subject 4, 8, 9 of SSMVEP dataset, who have the bad classification performance, the model with attention mechanism has a higher accuracy growth. For subjects with high classification performance, such as subject 1, 9 of SSVEP dataset and subject 1, 10 of SSMVEP dataset, the accuracy of the model with attention mechanism is slightly reduced.

IV. CONCLUSION

In this paper, we have proposed an attention-based parallel multiscale convolutional neural network (AMS-CNN), for the classification of Visual Evoked Potentials EEG. In contrast to the existing methods, AMS-CNN achieves a higher classification accuracy and a smaller variance across subjects, especially under fatigue state. We employ two parallel convolutional layers with small and large temporal filters respectively, which can effectively extract robust temporal representations. Then, we apply two sequential convolution blocks for spatial fusion and temporal fusion to extract advanced feature representations. Further, we use attention mechanism to weight the features at different moments according to the output-related interest. For each feature, the attention model generate an independent group of values which would be used as weights. Finally, we employ a full connected layer with softmax activation function for classification. Results show that multiscale convolution and attention mechanism can effectively improve the classification accuracy, especially for the subjects with less obvious characteristics. All the above demonstrate that our proposed framework can serve as a powerful tool for improving the classification performance of Visual Evoked Potentials EEG signals.

REFERENCES

- [1] J. Zhao, W. Li, X. Q. Mao, H. Hu, L. W. Niu, and G. S. Chen, "Behavior-based SSVEP hierarchical architecture for telepresence control of humanoid robot to achieve full-body movement," *IEEE Trans. Cogn. Dev. Syst.*, vol. 9, no. 2, pp. 197–209, Jun. 2017.

- [2] C. T. Lin *et al.*, "A wireless multifunctional SSVEP-based brain-computer interface assistive system," *IEEE Trans. Cogn. Dev. Syst.*, vol. 11, no. 3, pp. 375–383, Sep. 2019.
- [3] H. Strasburger, W. Scheidler, and I. Rentschler, "Amplitude and phase characteristics of the steady-state visual evoked potential," *Appl. Opt.*, vol. 27, no. 6, pp. 1069–88, Mar. 1988.
- [4] S. Ge, R. M. Wang, Y. Leng, H. X. Wang, P. Lin and K. Iramina, "A double-partial least-squares model for the detection of steady-state visual evoked potentials," *IEEE J. Biomed. Health Inform.*, vol. 21, no. 4, pp. 897–903, Jul. 2017.
- [5] J. Xie, G. H. Xu, J. Wang, F. Zhang, and Y. Z. Zhang, "Steady-state motion visual evoked potentials produced by oscillating newton's rings: Implications for brain-computer interfaces," *PLoS One*, vol. 7, Jun. 2012, Art no. e39707.
- [6] W. Q. Yan, G. H. Xu, J. Xie, M. Li, and Z. Y. Dan, "Four novel motion paradigms based on steady-state motion visual evoked potential," *IEEE Trans. Biomed. Eng.*, vol. 65, no. 8, pp. 1696–1704, Aug. 2018.
- [7] Z. K. Gao *et al.*, "An adaptive optimal-kernel time-frequency representation-based complex network method for characterizing fatigued behavior using the SSVEP-based BCI system," *Knowl.-Based Syst.*, vol. 152, pp. 163–171, Jul. 2018.
- [8] S. Ajami, A. Mahnam, and V. Abotalebi, "An adaptive SSVEP-based brain-computer interface to compensate fatigue-induced decline of performance in practical application," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 26, no. 11, pp. 2200–2209, Nov. 2018.
- [9] Z. L. Lin, C. S. Zhang, W. Wu, and X. R. Gao, "Frequency recognition based on canonical correlation analysis for SSVEP-based BCIs," *IEEE Trans. Biomed. Eng.*, vol. 53, pp. 2610–2614, Dec. 2006.
- [10] Y. Zhang, G. X. Zhou, J. Jin, M. J. Wang, X. Y. Wang, and A. Cichocki, "L1-regularized multiway canonical correlation analysis for SSVEP-Based BCI," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 21, no. 6, pp. 887–896, Nov. 2013.
- [11] X. G. Chen, Y. J. Wang, S. K. Gao, T. P. Jung, and X. R. Gao, "Filter bank canonical correlation analysis for implementing a high-speed SSVEP-based brain-computer interface," *J. Neural Eng.*, vol. 12, Aug. 2015, Art no. 046008.
- [12] H. Vu, B. Koo, and S. Choi, "Frequency detection for SSVEP-based bci using deep canonical correlation analysis," in *Proc. IEEE Int. Conf. Syst., Man, Cybern.*, 2016, pp. 1983–1987.
- [13] Z. Zhang, C. C. Wang, K. K. Ang, A. A. P. Wai and C. T. Guan, "Spectrum and phase adaptive cca for SSVEP-based brain computer interface," *2018 40th Annu. Int. Conf. IEEE Eng. Med. Biol. Soc., IEEE Eng. Med. Biology Soc. Conf. Proc.*, vol. 2018, pp. 311–314, Jul. 2018.
- [14] S. E. Nwachukwu *et al.*, "An SSVEP recognition method by combining individual template with CCA," *Proc. 3rd Int. Conf. Innov. Artif. Intell.*, 2019, pp. 6–10.
- [15] Y. LeCun, Y. Bengio, and G. Hinton, "Deep learning," *Nature*, vol. 521, no. 7553, pp. 436–444, May 2015.
- [16] L. Y. Jing, M. Zhao, P. Li, and X. Q. Xu, "A convolutional neural network based feature learning and fault diagnosis method for the condition monitoring of gearbox," *Measurement*, vol. 111, pp. 1–10, Dec. 2017.
- [17] H. Lee, M. Jung, and J. Tani, "Recognition of visually perceived compositional human actions by multiple spatio-temporal scales recurrent neural networks," *IEEE Trans. Cogn. Dev. Syst.*, vol. 10, no. 4, pp. 1058–1069, Dec. 2018.
- [18] Z. H. Cui, F. Xue, X. J. Cai, Y. Cao, G. G. Wang and J. J. Chen, "Detection of malicious code variants based on deep learning," *IEEE Trans. Ind. Inform.*, vol. 14, no. 7, pp. 3187–3196, Jul. 2018.
- [19] Q. C. Zhang, L. T. Yang, Z. Yan, Z. K. Chen, and P. Li, "An efficient deep learning model to predict cloud workload for industry informatics," *IEEE Trans. Ind. Inform.*, vol. 14, no. 7, pp. 3170–3178, Jul. 2018.
- [20] W. D. Dang, Z. K. Gao, L. H. Hou, D. M. Lv, S. M. Qiu, and G. R. Chen, "A novel deep learning framework for industrial multiphase flow characterization," *IEEE Trans. Ind. Inform.*, vol. 15, no. 11, pp. 5954–5962, Nov. 2019.
- [21] Z. K. Gao, W. D. Dang, M. X. Liu, W. Guo, K. Ma, and G. R. Chen, "Classification of EEG signals on VEP-based BCI systems with broad learning," *IEEE Trans. Syst. Man Cybern. Syst.*, to be published, doi: 10.1109/TSMC.2020.2964684.
- [22] H. Cecotti and A. Graser, "Convolutional neural networks for P300 detection with application to brain-computer interfaces," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 33, no. 3, pp. 433–445, Mar. 2011.
- [23] A. Antoniadis *et al.*, "Detection of interictal discharges with convolutional neural networks using discrete ordered multichannel intracranial EEG," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 25, no. 12, pp. 2285–2294, Dec. 2017.
- [24] P. B. Zhang, X. Wang, W. H. Zhang and J. F. Chen, "Learning spatial-spectral-temporal EEG features with recurrent 3D convolutional neural networks for cross-task mental workload assessment," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 27, no. 1, pp. 31–42, Jan. 2019.
- [25] X. Zhang and D. R. Wu, "On the vulnerability of CNN classifiers in EEG-based BCIs," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 27, no. 5, pp. 814–825, May 2019.
- [26] D. Ahmedt Aristizabal *et al.*, "Identification of children at risk of schizophrenia via deep learning and EEG responses," *IEEE J. Biomed. Health Inform.*, vol. 25, no. 1, pp. 69–76, Jan. 2021.
- [27] Y. Zhang, Y. Guo, P. Yang, W. Chen and B. Lo, "Epilepsy seizure prediction on EEG using common spatial pattern and convolutional neural network," *IEEE J. Biomed. Health Inform.*, vol. 24, no. 2, pp. 465–474, Feb. 2020.
- [28] U. R. Acharya, S. L. Oh, Y. Hagiwara, J. H. Tan, and H. Adeli, "Deep convolutional neural network for the automated detection and diagnosis of seizure using EEG signals," *Comput. Biol. Med.*, vol. 100, pp. 270–278, Sep. 2018.
- [29] J. X. Chen, P. W. Zhang, Z. J. Mao, Y. F. Huang, D. M. Jiang, and A. N. Zhang, "Accurate EEG-based emotion recognition on combined features using deep convolutional neural networks," *IEEE Access*, vol. 7, pp. 44317–44328, 2019.
- [30] Z. K. Gao *et al.*, "EEG-based spatio-temporal convolutional neural network for driver fatigue evaluation," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 30, no. 9, pp. 2755–2763, Sep. 2019.
- [31] N. S. Kwak, K. R. Muller, and S. W. Lee, "A convolutional neural network for steady state visual evoked potential classification under ambulatory environment," *PLoS One*, vol. 12, Feb. 2017, Art no. e0172578.
- [32] X. Zhang *et al.*, "A convolutional neural network for the detection of asynchronous steady state motion visual evoked potential," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 27, no. 6, pp. 1303–1311, Jun. 2019.
- [33] N. Waytowich *et al.*, "Compact convolutional neural networks for classification of asynchronous steady-state visual evoked potentials," *J. Neural Eng.*, vol. 15, Dec. 2018, Art no. 066031.
- [34] G. H. Dai, J. Zhou, J. H. Huang, and N. Wang, "HS-CNN: A CNN with hybrid convolution scale for EEG motor imagery classification," *J. Neural Eng.*, vol. 17, no. 1, Feb. 2020, Art no. 016025.
- [35] Y. Li, X. R. Zhang, B. Zhang, M. Y. Lei, W. G. Cui, and Y. Z. Guo, "A channel-projection mixed-scale convolutional neural network for motor imagery EEG decoding," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 27, no. 6, pp. 1170–1180, Jun. 2019.
- [36] D. L. Zhang, K. X. Chen, D. B. Jian, and L. N. Yao, "Motor imagery classification via temporal attention cues of graph embedded EEG signals," *IEEE J. Biomed. Health Inform.*, vol. 24, no. 9, pp. 2570–2579, Sep. 2020.
- [37] W. Qu *et al.*, "A residual based attention model for EEG based sleep staging," *IEEE J. Biomed. Health Inform.*, vol. 24, no. 10, pp. 2833–2843, Oct. 2020.
- [38] A. Supratak, H. Dong, C. Wu, and Y. K. Guo, "DeepSleepNet: A model for automatic sleep stage scoring based on raw single-channel EEG," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 25, no. 11, pp. 1998–2008, Nov. 2017.
- [39] Q. H. Yao, R. X. Wang, X. M. Fan, J. K. Liu, and Y. Li, "Multi-class arrhythmia detection from 12-lead varied-length ECG using attention-based time-incremental convolutional neural network," *Inf. Fusion*, vol. 53, pp. 174–182, Jan. 2020.
- [40] A. Delorme and S. Makeig, "EEGLAB: An open source toolbox for analysis of single-trial EEG dynamics including independent component analysis," *J. Neurosci. Methods*, vol. 134, pp. 9–21, Mar. 2004.
- [41] S. Zeki, J. D. Watson, C. J. Lueck, K. J. Friston, C. Kennard, and R. S. Frackowiak, "A direct demonstration of functional specialization in human visual cortex," *J. Neurosci.*, vol. 11, no. 3, pp. 641–649, Mar. 1991.